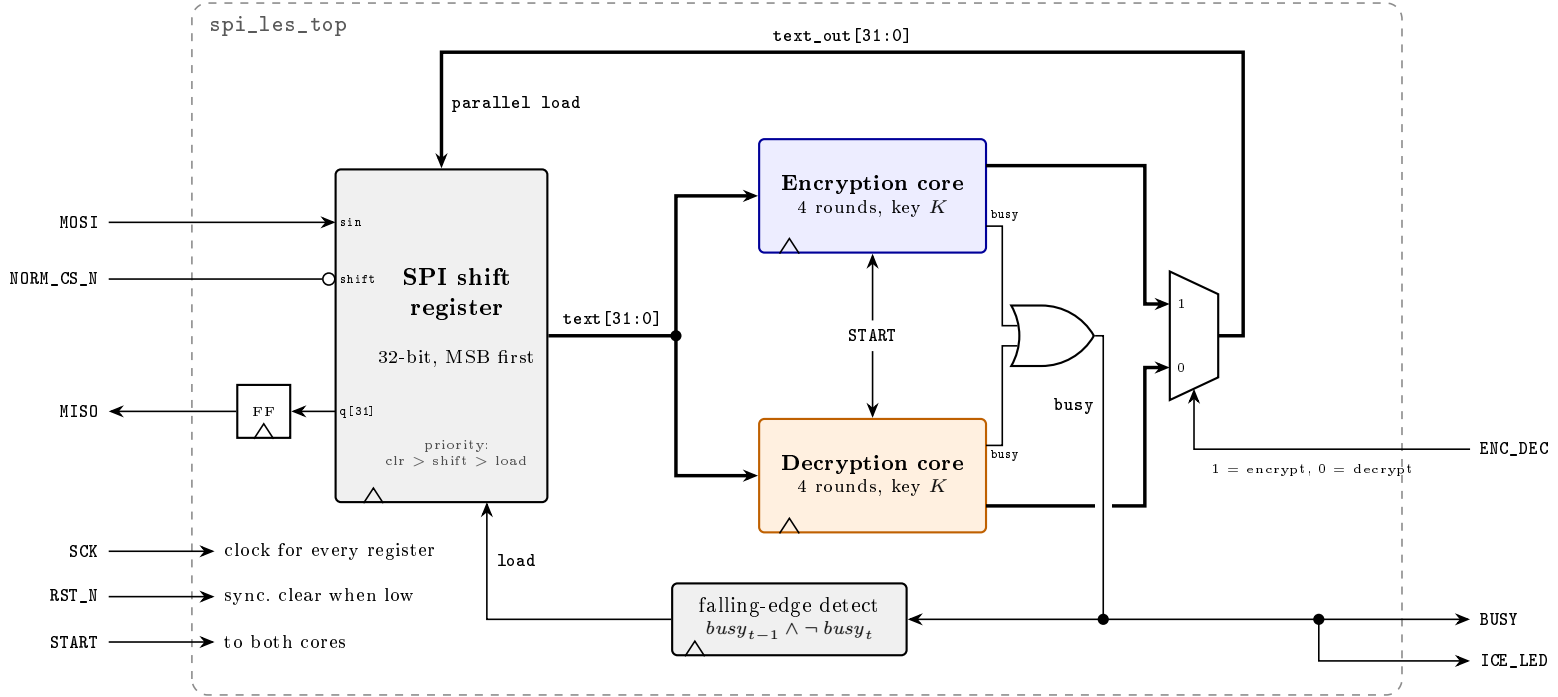
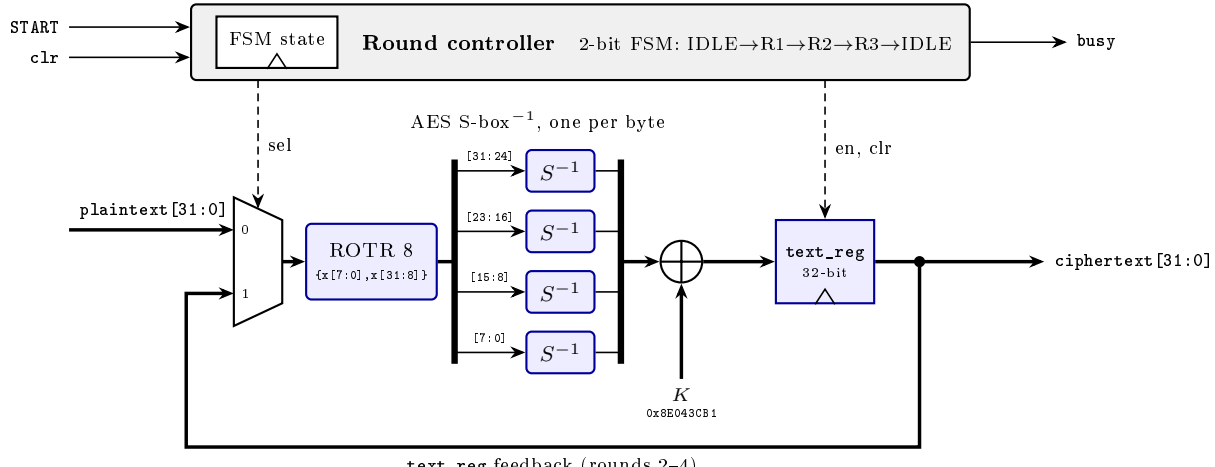
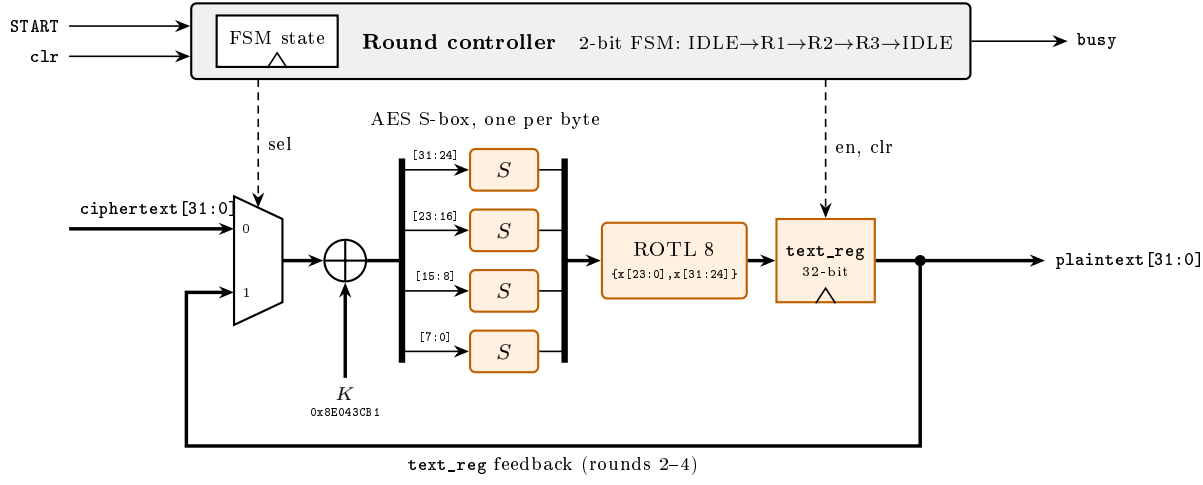

spi_les_top
text_out[31:0]
parallel load
MOSI
NORM_CS_N
MISO
sin
shift
SPI shift register
32-bit, MSB first
FF
q[31]
priority:
clr > shift > load
text[31:0]
Encryption core
4 rounds, key K
Decryption core
4 rounds, key K
START
busy
busy
busy
1
0
ENC_DEC
1 = encrypt, 0 = decrypt
SCK
clock for every register
RST_N
sync. clear when low
START
to both cores
load
falling-edge detect
$busy_{t-1} \wedge \neg\, busy_t$
BUSY
ICE_LED

Encryption core



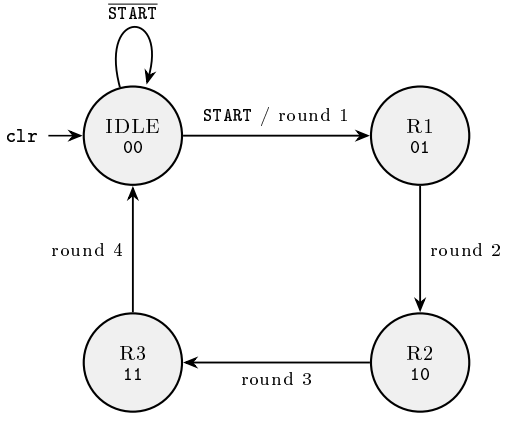
$$x_0 = \text{plaintext}, \qquad x_{i+1} = S^{-1}(x_i \ggg 8) \oplus K, \qquad \text{ciphertext} = x_4$$

Decryption core



$$y_0 = \text{ciphertext}, \qquad y_{i+1} = S(y_i \oplus K) \lll 8, \qquad \text{plaintext} = y_4$$

Round controller (identical in both cores)



state	sel	en	busy	text_reg on next SCK edge	
IDLE, START= 0	0	0	0	hold	
IDLE, START= 1	0	1	1	$\leftarrow R$(input)	round 1
R1	1	1	1	$\leftarrow R$(text_reg)	round 2
R2	1	1	1	$\leftarrow R$(text_reg)	round 3
R3	1	1	1	$\leftarrow R$(text_reg)	round 4

busy is combinational: it rises as soon as START is seen in IDLE and falls after round 4. R is the core's round function. The FSM state register only sequences the rounds; the data is held in the separate 32-bit text_reg. clr returns the FSM to IDLE and zeroes text_reg.